

A LOCALIZATION STUDY OF THE CYTOCHROME *P*-450₂₁-LINKED MONOOXYGENASE SYSTEM IN ADULT RAT BRAIN

KAZUHIKO IWAHASHI,^{1*} YOSHINORI KAWAI,² HIROSHI SUWAKI,³ KIYOSHI HOSOKAWA³ and YOSHIYUKI ICHIKAWA¹

¹Department of Biochemistry, ²Department of Anatomy and ³Department of Neuropsychiatry, Kagawa Medical School, Kagawa 761-07, Japan

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Summary—Immunohistochemical studies were performed to investigate the localization of the cytochrome *P*-450₂₁-linked monooxygenase system in rat brain using a specific antibody against bovine adrenal cytochrome *P*-450₂₁ (*P*-450XXIA1), which was purified electrophoretically as a single protein band and with a heme content of 18.0 nmol/mg protein from adrenocortical microsomes. The cytochrome *P*-450₂₁ was found to be mainly localized in the tractus reticulothalamicus and other ascending fibers in adult rat brains. This finding indicated that deoxycorticosterone or its derivatives could be implicated in the regulation of consciousness and the induction of an anesthetic effect.

INTRODUCTION

Most steroid hormones are synthesized by cytochrome *P*-450-linked monooxygenase systems in microsomes and mitochondria of adrenocortices, ovaries, testes, placenta and brain [1–3]. The cytochrome *P*-450-linked monooxygenase system of the microsomal type is composed of two components: NADPH-cytochrome *P*-450 reductase (E.C. 1.6.2.4) and cytochrome *P*-450. The cytochrome *P*-450₂₁-linked monooxygenase system (steroid 21-monooxygenase) is also of the microsomal type, and catalyzes the steroid 21-hydroxylations which produce deoxycorticosterone (DOC) from progesterone, and 11-deoxycortisol from 17-hydroxyprogesterone [3]. Cytochrome *P*-450₂₁ (*P*-450XXIA1) is well known to be localized not only in adrenocortical microsomes, but also in extra-adrenal tissue microsomes [4]. We have already reported the localization of cytochrome *P*-450_{sc} in the brain [5], it being the first step enzyme for the biosynthesis of steroid hormones.

In this paper, to investigate the physiological function of the cytochrome *P*-450 involved in

steroid biosynthesis in the central nervous system, the localization of cytochrome *P*-450₂₁ in the brains of adult rats was determined by means of immunohistochemical techniques using a specific antibody against cytochrome *P*-450₂₁, and the activity of the steroid 21-monooxygenase system in the absence and presence of the antibody was also examined. A preliminary account of the localization of cytochrome *P*-450₂₁ in the brain has been presented as a selected paper of the Japan Brain Research Association [6].

MATERIALS AND METHODS

Animals

Female and male Wistar rats (body wt, 100–350 g; age, 5–20 weeks) bred in the laboratory were used throughout. Female Japanese white rabbits (body wt, 1.5 kg; age, 6 months) were subjected to bovine cytochrome *P*-450₂₁ immunization.

Purification of cytochrome P-450₂₁

Cytochrome *P*-450₂₁ was purified from bovine adrenocortical microsomes by the method of Hiwatashi and Ichikawa [3]. The cytochrome *P*-450₂₁ gave a single protein band on SDS-polyacrylamide gel electrophoresis and its optical spectrum showed it to be optically

*Present address and address for correspondence: Department of Neuropsychiatry, Kagawa Medical School, Kagawa 761-07, Japan.

Abbreviations: SDS, sodium dodecyl sulfate; DAB, 3,3'-diaminobenzidine tetrahydrochloride; *P*-450₂₁, cytochrome *P*-450₂₁ (steroid 21-monooxygenase); G-6-P, glucose-6-phosphate.

highly pure. The specific heme content was 18.5 nmol/mg protein.

Preparation of microsomes from rat tissues

Rats (20 males) were killed by decapitation. Their brains and adrenal glands were immediately excised, and the brainstems were removed from the brains. Microsomes were prepared from the adrenal glands and brainstems by the methods of Rosenthal and Narasimhulu, respectively [7].

Preparation of an antibody against cytochrome P-450₂₁

Cytochrome P-450₂₁, which had been purified as described above, was used as an antigen. Female adult rabbits were immunized, and after the antibody against cytochrome P-450₂₁ had been prepared by precipitation with 50% saturated ammonium sulfate (pH 7.0), it was purified by the method of Pfeider *et al.* [8]. The specificity of the antibody was confirmed by screening against a homogeneous preparation of cytochrome P-450₂₁, on the basis of the following criteria: inhibition of the activity of the cytochrome P-450₂₁-linked monooxygenase system; the results of Ouchterlony double diffusion analysis; and no cross-reactivity with either cytochrome P-450_{11 β} (P-450XIB1) or cytochrome P-450_{sec} (P-450XIA1) [5]. The antibody against NADPH-cytochrome P-450 reductase was a generous gift from Dr M. Tsubaki (Himeji Institute of Technology, Hyogo, Japan). The reductase was purified from bovine liver microsomes and then injected in to female adult rabbits. The antibody was prepared as described above.

Cytochrome P-450₂₁-coupled Sepharose 4B affinity chromatography

Four milligrams of purified cytochrome P-450₂₁ was coupled with 4 ml of Sepharose 4B activated with cyanogen bromide (Pharmacia, Tokyo, Japan) [5]. Five milligrams of the antibody against cytochrome P-450₂₁ was applied to examine the immunoactivity on the affinity column, and the pass-through fraction was used as the nonspecific antibody as to cytochrome P-450₂₁ for control staining.

Preparation of a preimmune antibody

A preimmune antibody was prepared from serum of untreated female rabbits by a similar method to in the case of the antibody against cytochrome P-450₂₁.

Specificity of the antibody against cytochrome P-450₂₁

In order to demonstrate the specificity of the immunohistochemical staining with the antibody against cytochrome P-450₂₁, control staining of adjacent sections on the same side of the rat brain was performed. The latter involved the preimmune antibody, dilution of the specific antibody to extinction of the staining, and absorption of the antibody against cytochrome P-450₂₁ on antigen-coupled Sepharose 4B gel. The staining by the antibody was confirmed to be specific to cytochrome P-450₂₁ in sections of rat adrenocortex.

Immunohistochemistry

For the immunohistochemical study, 5 rats were perfused transcardially with 50 to 100 ml of 10 mM potassium phosphate buffer (pH 7.4) containing 0.9% (w/v) NaCl (buffer A) at room temperature, over 5 to 10 min, followed by 200–500 ml of 10% (v/v) paraformaldehyde in buffer A for 30 min. Brain coronal sections (20 μ m) were cut at the level of the superior colliculus and fixed for 24 h in Carnoy's solution at room temperature. Endogenous peroxidase was blocked by immersing the sections in a solution of 0.1% (v/v) hydrogen peroxide in methanol for 20 min.

The localization of cytochrome P-450₂₁ in the rat adrenocortex and brain was demonstrated by the avidin-biotin-peroxidase complex method of Hu *et al.* [9]. The peroxidase activity was detected with 3,3'-diamino-benzidine tetrahydrochloride (DAB), 0.5 mg/ml, in the presence of 0.01% (v/v) H₂O₂ in 50 mM Tris-HCl buffer (pH 7.6). After the sections had been rinsed, dehydrated and mounted, the slides were examined under a Nikon UFK-II microscope.

Chemicals

[1,2-³H]Progesterone was obtained from Amersham (Tokyo, Japan). The radioactivity was determined with a Packard Tri-Carb liquid scintillator. Avidin-biotin-peroxidase complex was obtained from Vectastain Labs (Burlingame, CA, U.S.A., ABC Kit). Sepharose 4B activated with cyanogen bromide was purchased from Pharmacia, and NADPH, G-6-P and G-6-P dehydrogenase were from Oriental Yeast Co., Ltd (Tokyo, Japan). All other chemicals were of the highest quality commercially available.

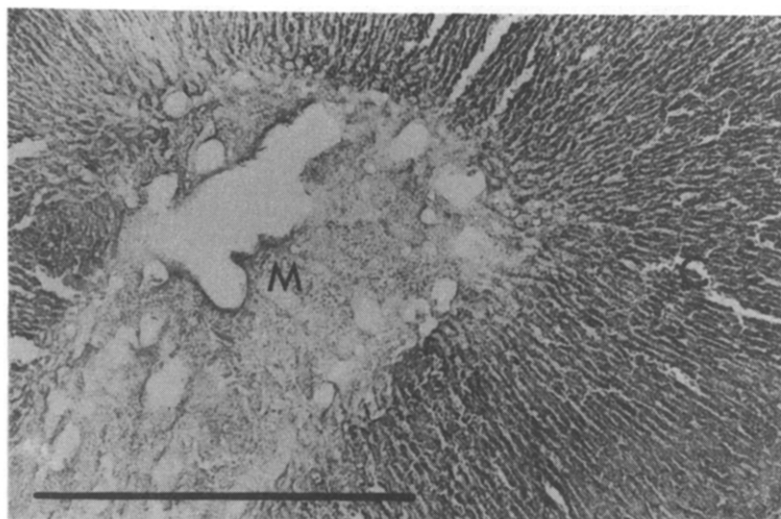


Fig. 1. Microphotograph showing immunoperoxidase staining of a section of a rat adrenal gland with the antibody against cytochrome *P*-450₂₁. All parenchymal cells in the cortex are intensely immunostained. The cells in the adrenomedulla are totally unstained. C, adrenocortex; M, adrenomedulla; bar = 1 mm.

Enzyme activity

[³H]Progesterone (2 μ M, 1 μ Ci) was incubated with 1 mg of microsomes, and a NADPH generating system comprising 5 mM G-6-P and 0.5 U of G-6-P dehydrogenase in 50 mM K-phosphate buffer, pH 7.4, in a total volume of 1 ml, and the enzymatic reaction was started by the addition of 200 μ M NADPH and then performed for 20 min at 37°C.

After the reaction, the metabolized steroids were extracted with CH₂Cl₂. The extracted steroids were analyzed by thin layer chromatography (benzene-acetate, 3:1, v/v), and the radioactivity of the deoxycorticosterone spot was measured as the progesterone 21-monooxygenase activity by the method of Hiwatashi and Ichikawa [10].

Measurement of radioactivity

Radioactivity was determined with a Packard TriCarb liquid scintillation spectrometer, Model 3320 (Packard Instrument Co. Inc.). Brays solution was used as a liquid scintillator [11].

RESULTS

The localization of cytochrome *P*-450₂₁ (steroid 21-monooxygenase) and its NADPH-cytochrome *P*-450 reductase in adult rat brain was examined immunohistochemically using antibodies against cytochrome *P*-450₂₁ and its reductase. The specificity of the antibody against cytochrome *P*-450₂₁ was confirmed by its staining of sections of rat adrenal glands.

The adrenocortical cells were observed to be stained effectively by various concentrations of the antibody against cytochrome *P*-450₂₁ in 20 min at 25°C, while the adrenal medullary cells did not exhibit even weak positive staining with the antibody (Fig. 1). The adrenocortical cells could not be stained with the preimmune antibody or the antibody pretreated by cytochrome *P*-450₂₁-affinity chromatography. Therefore, we judged that the antibody against cytochrome *P*-450₂₁ can be used effectively for the determination of the tissue localization of cytochrome *P*-450₂₁ in other rat tissues. Accordingly, the antibody against cytochrome *P*-450₂₁ was applied immunohistochemically to investigate the localization of cytochrome *P*-450₂₁ in rat brain sections. Figure 2A shows a microphotograph of a stained section at the level of the superior colliculus of an adult rat brain. The intensely stained areas are illustrated as dotted areas in Fig. 2B. Area (a) contains mainly the ascending sensory fibers of the tractus reticulothalamicus, and area (b) comprises the pontine reticular regions that contain the cells and ascending fibers of the tractus reticulothalamicus. The immunostaining of a brain section with the treated antibody, which was pretreated by affinity chromatography on cytochrome *P*-450₂₁-coupled Sepharose 4B, was negative or only very weakly positive, even with a high concentration of the antibody of up to 100 μ g protein/ml under the same experimental conditions (Fig. 2C). Figure 3 shows a magnified microphotograph of a stained section at the level of the superior colliculus of adult rat brain.

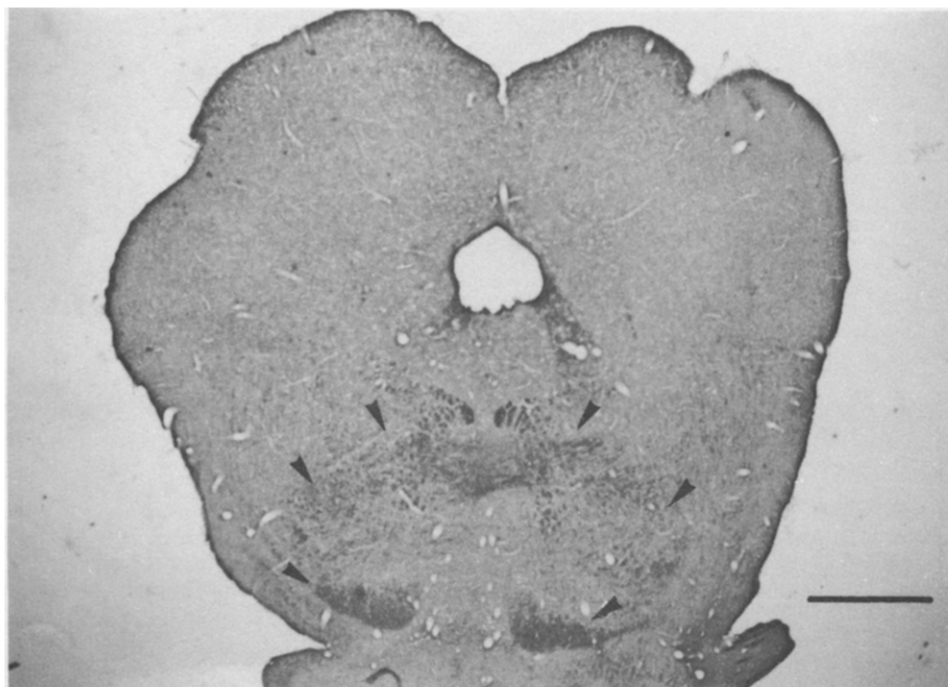


Fig. 2A. Microphotograph showing immunoperoxidase staining of a coronal section of a rat brain with the antibody against cytochrome $P-450_{21}$. Brain coronal sections were cut at the level of the superior colliculus. Clearly stained cells and ascending fibers can be seen mainly in the tractus reticulothalamicus. Original magnification, $\times 20$, bar = 1 mm.

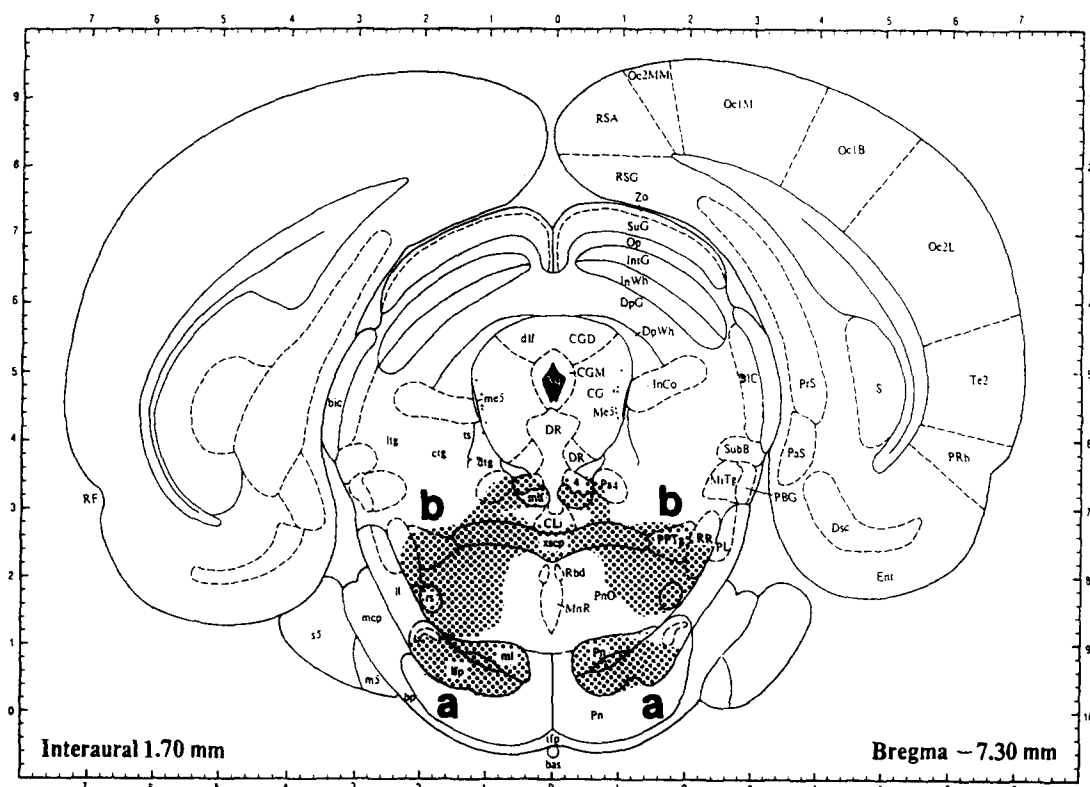


Fig. 2B. Illustration of a microphotograph of a coronal section of a rat brain. The brain section was cut at the level of the superior colliculus. This illustration is taken from Ref. [12, Fig. 47]. The dotted areas show the stained areas in Fig. 2A. lfp, longitudinal fasciculus pons; ml, medial lemniscus; pn, pontine nuclei; pno, pontine reticular nu. oral; pptg, pedunculopontine tegmental nu; rr, retrorubral nu; xscp, decussation sup Cb peduncle.

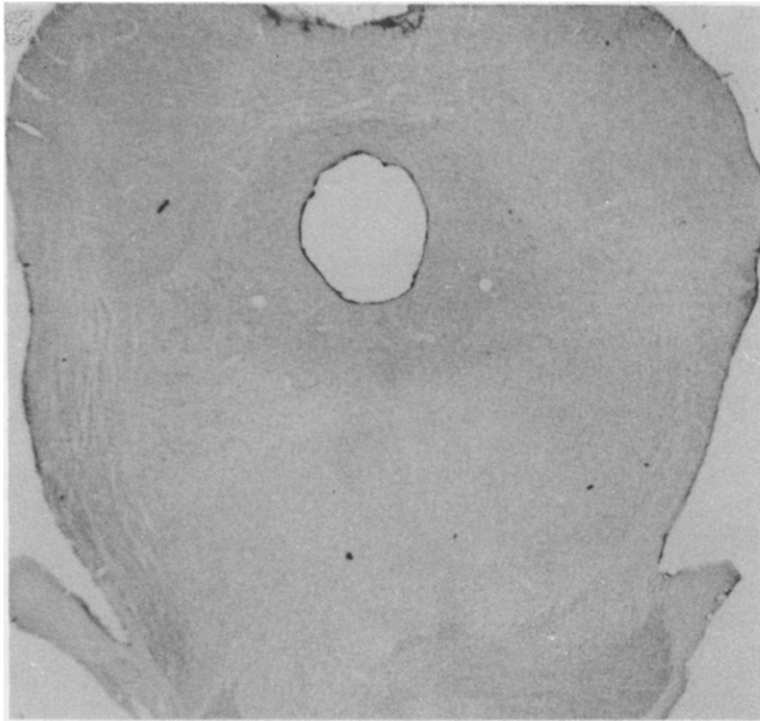


Fig. 2C. Microphotograph of a coronal section at the level of the superior colliculus of a rat brain showing staining with the preimmune antibody, which was obtained by affinity chromatography of cytochrome *P-450₂₁*, to remove the specific antibody against cytochrome *P-450₂₁*. Original magnification, $\times 20$; bar = 1 mm.

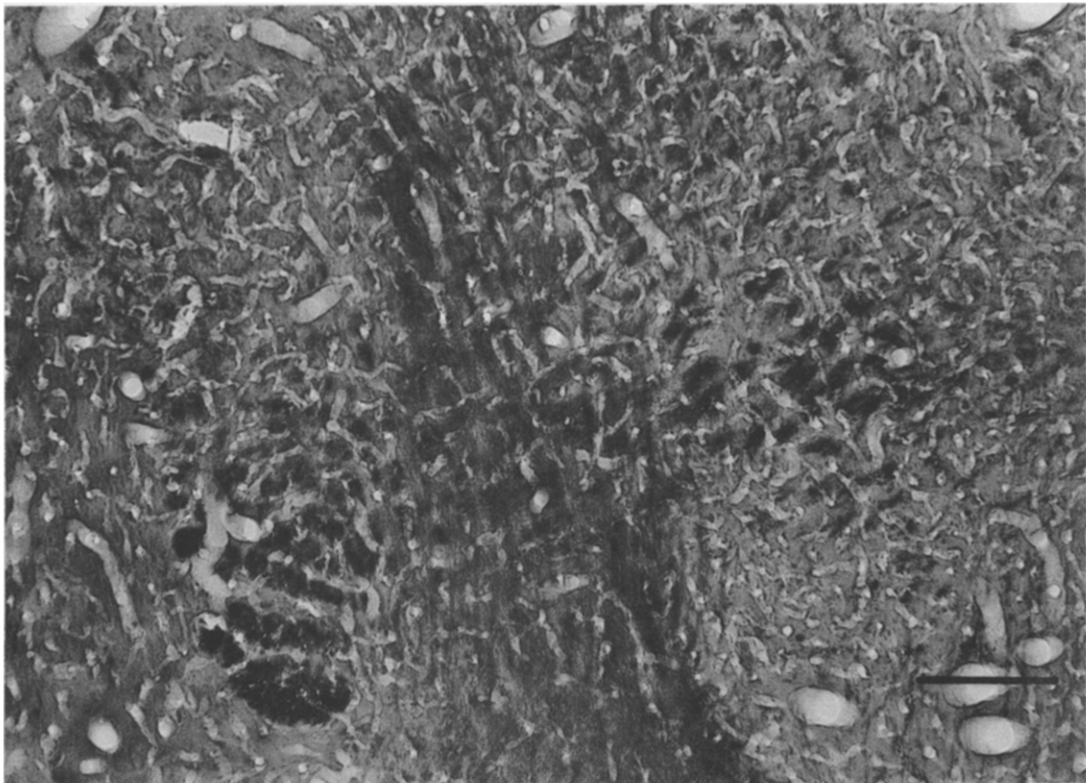


Fig. 3. Microphotograph showing immunoperoxidase staining of the magnified superior colliculus region of a rat brain with the antibody against cytochrome *P-450₂₁*. Brain coronal sections were cut at the level of the superior colliculus. Clearly stained cells or ascending fibers can be mainly seen in the tractus reticulothalamicus. Original magnification, $\times 320$; bar = 60 μ m.

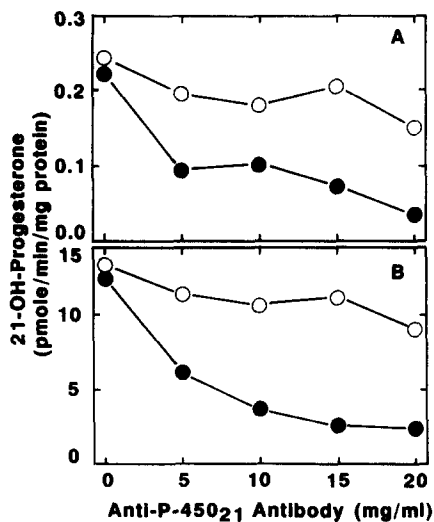


Fig. 4. Immunochemical inhibition of the activities of progesterone 21-monooxygenase systems in microsomes of male rat brainstems and adrenal glands. The effects of the antibody against cytochrome *P*-450₂₁ and the preimmune antibody on the activities are shown. (A) The activity of progesterone 21-monooxygenase of microsomes of brainstems of male rat brains was examined in the presence of the antibody against cytochrome *P*-450₂₁ or the preimmune antibody. (B) The activity of progesterone 21-monooxygenase of microsomes of male rat adrenal glands was examined under the same experimental conditions as in (A). Closed circles, antibody against cytochrome *P*-450₂₁; open circles, preimmune antibody.

There was no difference in the intensity of the immunostaining with the antibody against cytochrome *P*-450₂₁ between female and male adult rat brains. In addition, a difference in the immunostaining could not be observed between the right and left sides of rat female or male brains (data not shown).

The activity of progesterone 21-monooxygenation of the cytochrome *P*-450₂₁-linked monooxygenase system was detected in the microsomes of rat brainstems, the specific activity being 0.24 pmol/min/mg protein of the brainstem microsomes. The enzymatic activity could not be detected in the microsomes of the rat brain cortices. The activity of progesterone 21-monooxygenase in the brainstems was considerably weak in comparison with that in the adrenal glands. The activity in the brainstem microsomes was apparently one-fiftieth as much as that detected in the rat adrenal microsomes. We examined whether this weak activity was due to the cytochrome *P*-450₂₁-linked monooxygenase system or not. The activities of the progesterone 21-monooxygenases of the microsomes of the brainstem and adrenocortex were significantly inhibited by the antibody against cytochrome *P*-450₂₁ [Fig. 4(A and B)]. The

activity of the progesterone 21-monooxygenase in the brainstem microsomes was also inhibited significantly by the antibody against NADPH-cytochrome *P*-450 reductase (data not shown). These results demonstrate that the activity of the steroid 21-monooxygenase system was detected significantly in rat brainstems.

DISCUSSION

The immunohistochemical localization of cytochrome *P*-450₂₁ in the brains of adult rats was examined using a specific antibody against bovine adrenocortical cytochrome *P*-450₂₁, and it was found that mainly the neuronal cell, glial cell bodies and fibers in the tractus reticulothalamicus of the brain are selectively and specifically immunostained. In addition to this region, positive or weak positive staining of the fasciculus longitudinalis medialis and the neighboring sensory nerve fibers was also observed. These stained tissues comprised mainly nerve fibers.

The activity of progesterone 21-hydroxylation of the cytochrome *P*-450₂₁-linked monooxygenase system was detected in the brainstem microsomes. This activity supports the immunohistochemical finding that the selective and specific localization of cytochrome *P*-450₂₁ is mainly in the tractus reticulothalamicus of adult rat brain.

These results suggest the idea that deoxycorticosterone, the steroid hormone produced from progesterone, can be biosynthesized by the fibers neuronal cell and glial cell bodies of the tractus reticulothalamicus and other ascending fibers of the brain, and then that deoxycorticosterone or its steroidal derivatives will act as "neuroactive steroid" [12–15]. Both the physiological functions and action mechanisms of deoxycorticosterone or other steroids, which are biosynthesized in these regions of the brain are still poorly understood. However, it is possible that the neuroactive steroid probably has an effect on a brain region by binding to an as yet uncharacterized receptor, which is located intracellularly at the synapses [9].

We have reported on the localization of cytochrome *P*-450_{sec} (*P*-450XIA1) in rat brain [5], and it has been reported that steroid 5 α -reductase is present in the rat brain white matter [14, 16] and that aromatase (*P*-450XIX), which produces estrogen from androgen, is also localized in the preoptic area and hypothalamus of Japanese quail brain [17]. Although these

enzymatic activities are very weak in comparison to those of the adrenal glands, these results suggest that small amounts of various neuroactive steroid hormones are possibly synthesized from cholesterol in the adult rat brain.

In conclusion, this study has demonstrated that in the adult rat brain, the cytochrome P-450₂₁-linked monooxygenase system is localized in neuronal cells and/or glial cells surrounding nerve fibers, mainly in the tractus reticulothalamicus and other ascending fibers. Also, it is possible that deoxycorticosterone or its steroidal derivatives, which are mainly synthesized by the microsomal and mitochondrial cytochrome P-450-linked monooxygenase systems in adult rat brain, act as neuroactive steroid in the ascending reticular activating system, to maintain consciousness or, on the contrary, to induce an anesthetic effect.

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